CHM111 Practice Exam on chapter(s): 3, 4

1. $4\,Al(OH)_3 + 3\,C \rightarrow 6\,H_2O + 3\,CO_2 + 4\,Al$
2. 437. kg
3. 1.18 mL of iron (III) chloride
4. 2.05×10^1 g
5. 37.0%
6. 1.88×10^1 mol
7. 1.2×10^{49} g
8. $4\,O_2 + 3\,H_2 + N_2 \rightarrow 2\,H_2O + 2\,HNO_3$
9. 77%
10. Oxidized: Be Reduced: Pt
11. $2\,S + 3\,O_2 + 2\,H_2O \rightarrow 2\,H_2SO_4$
12. $2\,ZnS + 2\,C + 3\,O_2 \rightarrow 2\,CO + 2\,SO_2 + 2\,Zn$
13. $TiO_2 + 2\,Cl_2 + C + 2\,Mg \rightarrow Ti + CO_2 + 2\,MgCl_2$
14. $HNO_3(aq) + H_2O(\ell) \rightarrow NO_3^-(aq) + H_3O^+(aq)$
15. -1
16. 420 mM
17. $6\,H_2O + 5\,O_2 + P_4 \rightarrow 4\,H_3PO_4$
18. 1.24×10^{26} g
19. 7.94 g
20. 0.0214 L
21. 0.578 M
22. 0.073 g
23. $2\,LiOH + H_2CO_3 \rightarrow 2\,H_2O + Li_2CO_3$
24. 0.142 M
25. $Ni\,(s) + 2\,Au^+\,(aq) \rightarrow Ni^{2+}\,(aq) + 2\,Au\,(s)$
26. $C_6H_{12}O$
27. 31.7 g
28. 0.613 g
29. $2\,C + 2\,Cu_2S + 3\,O_2 \rightarrow 4\,Cu + 2\,SO_2 + 2\,CO$
30. H_3O^+, Br^-, and H_2O
31. 8.443×10^{-4} g
32. 8
33. A_2B_4
34. $0.69\ \frac{mol}{L}$
35. $Fe_2O_3 + 5\,CO \rightarrow 2\,Fe + 5\,CO_2$
36. $0.184\ \frac{mol}{L}$
37. $2\,RbBr + (H_3O)_2S \rightarrow 2\,H_3OBr + Rb_2S$
38. $101.96\ \frac{g}{mol}$
39. $6\,MnCO_3 + 8\,Al + 3\,O_2 \rightarrow 4\,Al_2O_3 + 6\,CO_2 + 6\,Mn$
40. $Cr^{3+}(aq) + PO_4^{3-}(aq) \rightarrow CrPO_4(s)$
41. 38.8%
42. 1.382×10^{21} sulfur atoms
43. Oxidizing Agent: CO_2 Reducing Agent: Li
44. $2\,K\,(s) + 2\,H_2O\,(l) \rightarrow H_2\,(g) + 2\,KOH\,(aq)$
45. 0.05518 moles
46. 3.9 mL
47. 11.07 g
48. 0.241 moles
49. 19.8 g
50. $6\,H_2O + C_3H_8 \rightarrow 10\,H_2 + 3\,CO_2$